

1: Student Academic Performance Dashboard

```
attendance <- c(90, 80, 70, 85, 95)
```

```
marks <- c(88, 75, 65, 80, 95)
```

```
plot(attendance, marks,
```

```
  main = "Student Academic Performance Dashboard",
```

```
  xlab = "Attendance (%)",
```

```
  ylab = "Internal Marks",
```

```
  pch = 19,
```

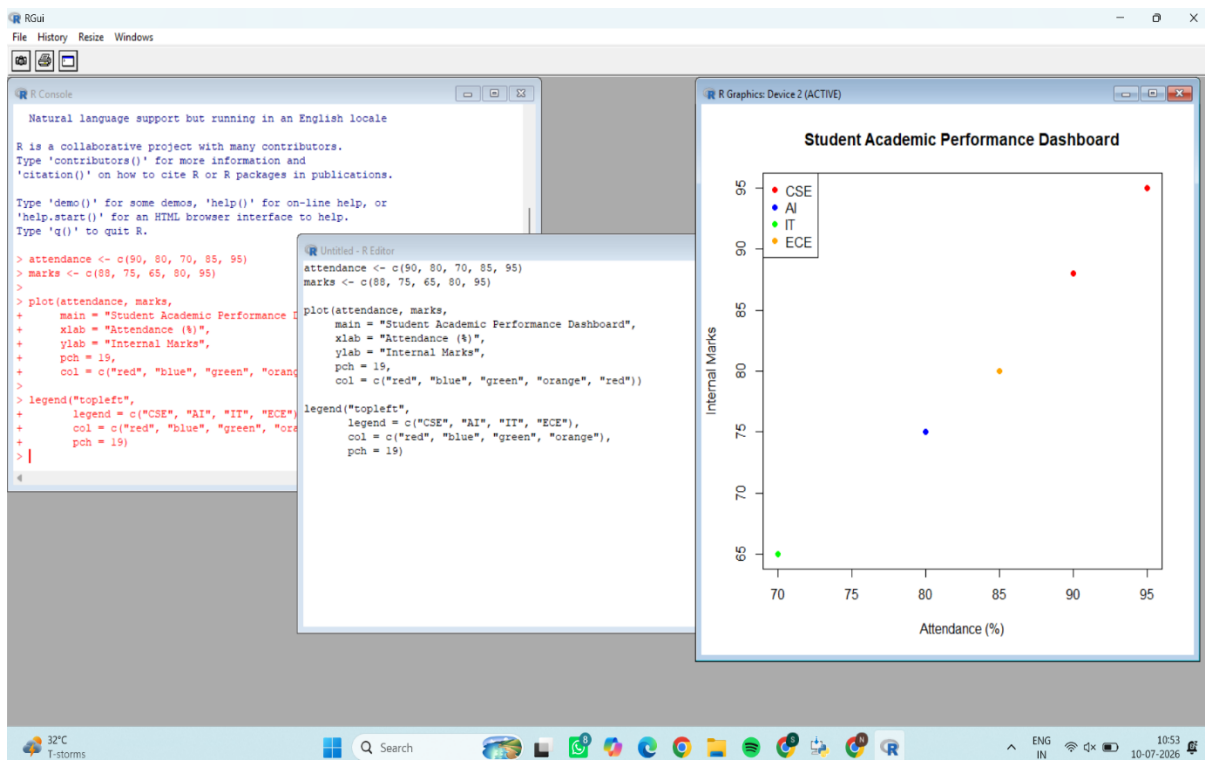
```
  col = c("red", "blue", "green", "orange", "red"))
```

```
legend("topleft",
```

```
  legend = c("CSE", "AI", "IT", "ECE"),
```

```
  col = c("red", "blue", "green", "orange"),
```

```
  pch = 19)
```



2: Electric Vehicle Market Analysis

```
battery <- c(30, 40, 50, 60, 75)
```

```
price <- c(10, 15, 20, 25, 35)
```

```
range <- c(200, 250, 300, 350, 450)
```

```
plot(battery, price,
```

```
main = "Electric Vehicle Market Analysis",
```

```
xlab = "Battery Capacity (kWh)",
```

```
ylab = "Price",
```

pch = 19,

$$\text{cex} = \text{range} / 100,$$

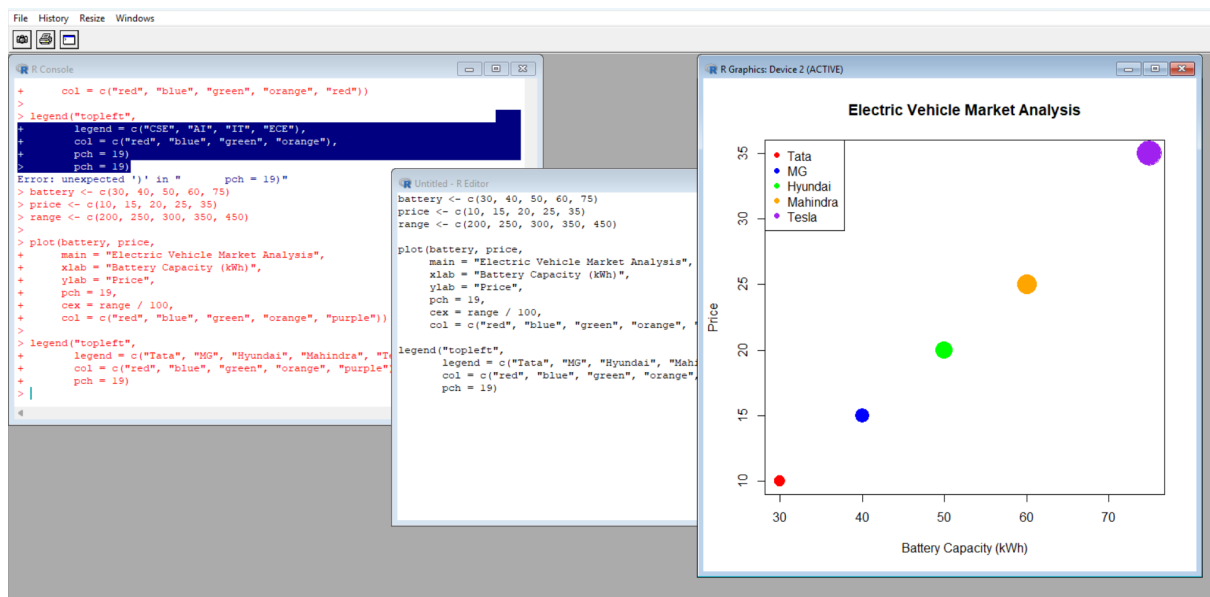
```
col = c("red", "blue", "green", "orange", "purple"))
```

```
legend("topleft",
```

```
legend = c("Tata", "MG", "Hyundai", "Mahindra", "Tesla"),
```

```
col = c("red", "blue", "green", "orange", "purple"),
```

pch = 19)



3: Hospital Patient Admission Analysis

```
department <- c("Cardiology", "Neurology", "Orthopedics", "Pediatrics")
```

```
patients <- c(120, 90, 150, 100)
```

```
cost <- c(5000, 7000, 6000, 4000)
```

```
bar <- barplot(patients,  
               names.arg = department,  
               col = c("red", "blue", "green", "orange"),  
               ylim = c(0, 170),  
               main = "Hospital Patient Admission Analysis",  
               xlab = "Department",  
               ylab = "Number of Patients")
```

```
text(bar, patients + 5, labels = cost)
```



4: Weather Monitoring System

```
date <- 1:5
```

```
chennai <- c(32, 33, 34, 33, 35)
```

```
delhi <- c(36, 37, 38, 37, 39)
```

```
mumbai <- c(30, 31, 31, 32, 33)
```

```
plot(date, chennai, type = "l",
```

```
col = "red", lty = 1, lwd = 2,
```

```
ylim = c(30, 40),
```

```
xlab = "Date",
```

```
ylab = "Temperature (°C)",
```

```
main = "Weather Monitoring System")
```

```
lines(date, delhi, col = "blue", lty = 2, lwd = 2)
```

```
lines(date, mumbai, col = "green", lty = 3, lwd = 2)
```

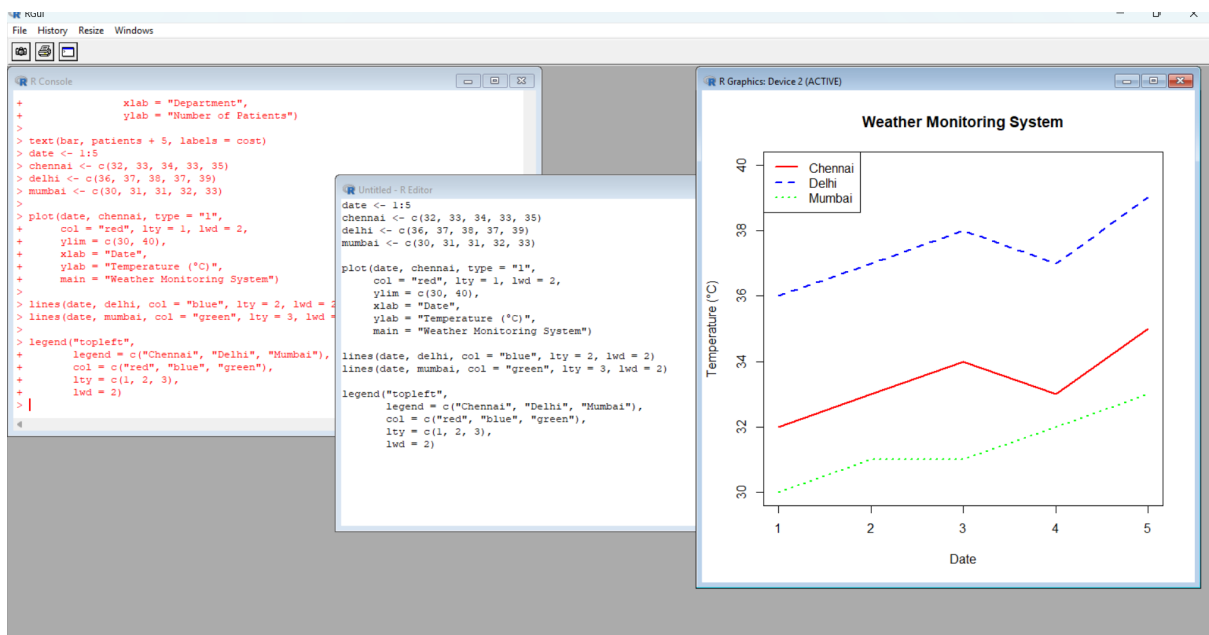
```
legend("topleft",
```

```
legend = c("Chennai", "Delhi", "Mumbai"),
```

```
col = c("red", "blue", "green"),
```

```
lty = c(1, 2, 3),
```

```
lwd = 2)
```

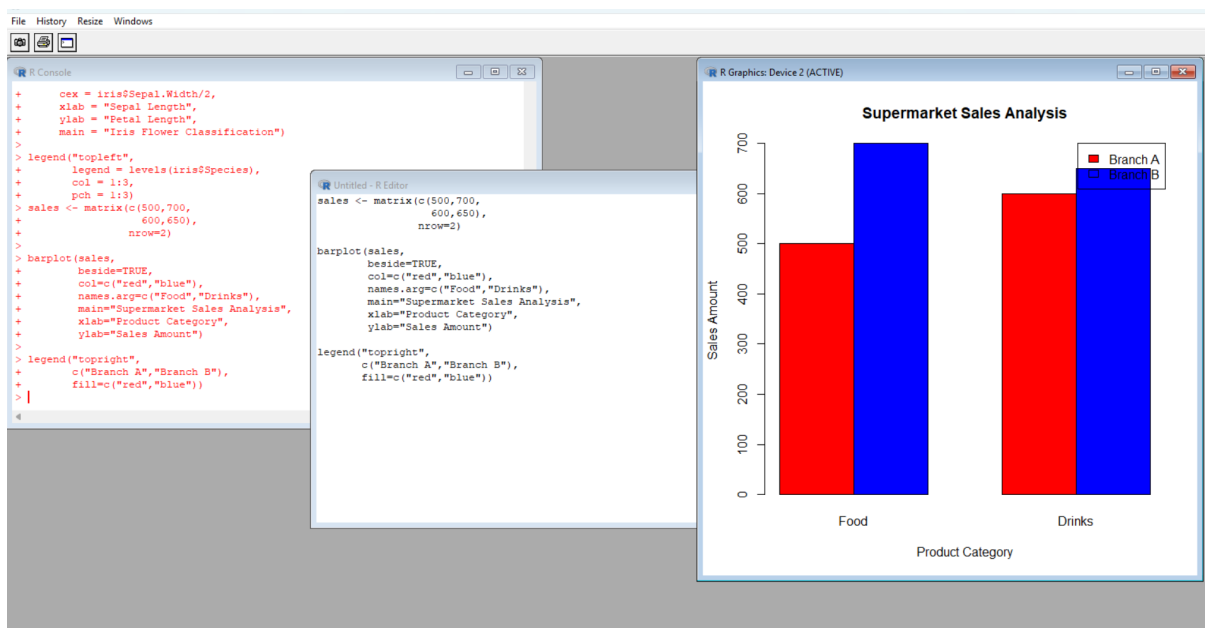


5: Supermarket Sales Analysis

```
sales <- matrix(c(500,700,  
                 600,650),  
               nrow=2)
```

```
barplot(sales,  
        beside=TRUE,  
        col=c("red","blue"),  
        names.arg=c("Food","Drinks"),  
        main="Supermarket Sales Analysis",  
        xlab="Product Category",  
        ylab="Sales Amount")
```

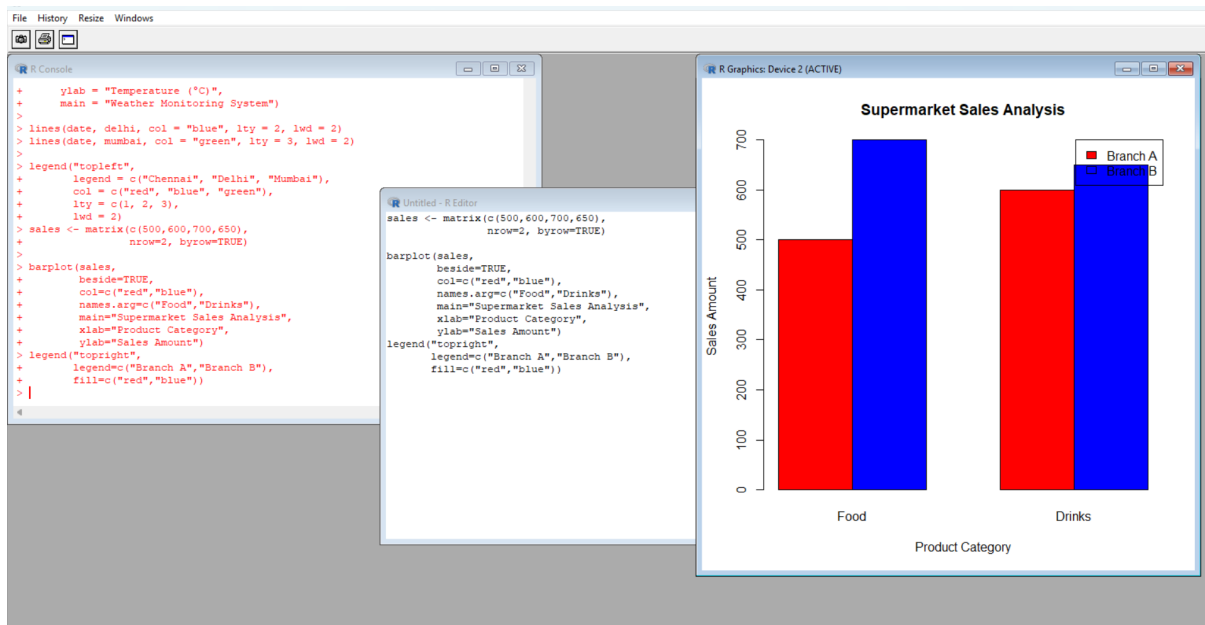
```
legend("topright",  
       c("Branch A","Branch B"),  
       fill=c("red","blue"))
```



6: Employee Salary Distribution

```
sales <- matrix(c(500,600,700,650),  
               nrow=2, byrow=TRUE)
```

```
barplot(sales,  
        beside=TRUE,  
        col=c("red","blue"),  
        names.arg=c("Food","Drinks"),  
        main="Supermarket Sales Analysis",  
        xlab="Product Category",  
        ylab="Sales Amount")  
legend("topright",  
       legend=c("Branch A","Branch B"),  
       fill=c("red","blue"))
```



7: Smartphone Market Comparison

```
ram <- c(4, 6, 8, 12)
```

```
price <- c(15000, 20000, 30000, 40000)
```

```
plot(ram, price,
```

```
  pch = c(15,16,17,18),
```

```
  cex = c(1,2,3,4),
```

```
  col = c("red","blue","green","orange"),
```

```
  xlab = "RAM",
```

```
  ylab = "Price",
```

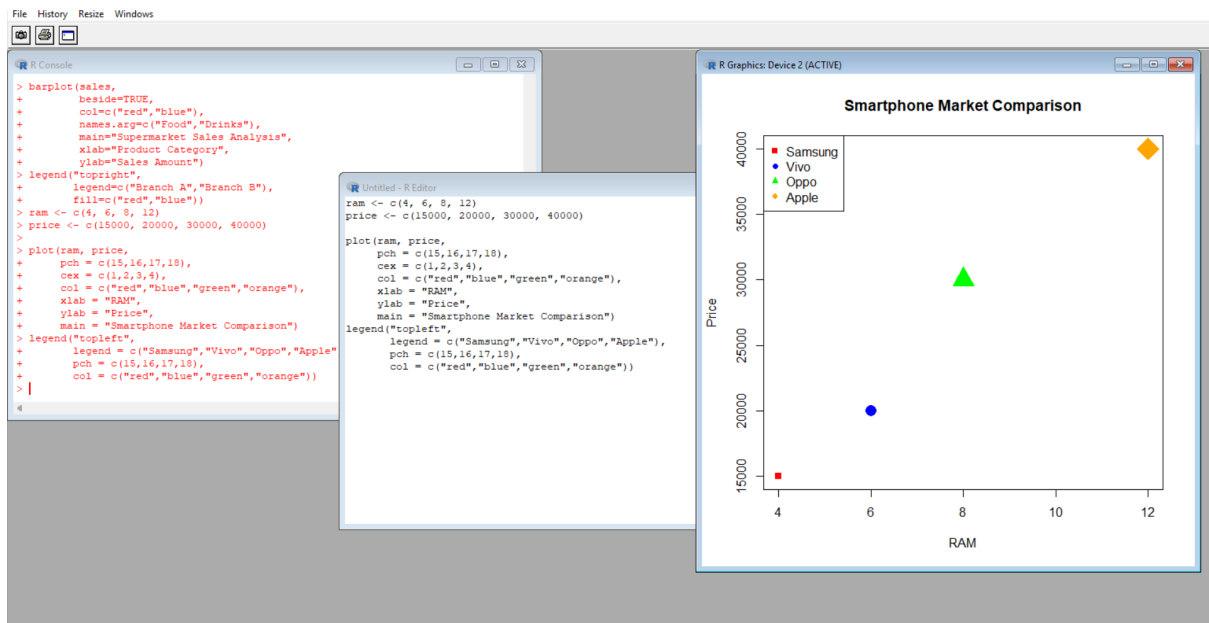
```
  main = "Smartphone Market Comparison")
```

```
legend("topleft",
```

```
  legend = c("Samsung","Vivo","Oppo","Apple"),
```

```
  pch = c(15,16,17,18),
```

```
  col = c("red","blue","green","orange"))
```

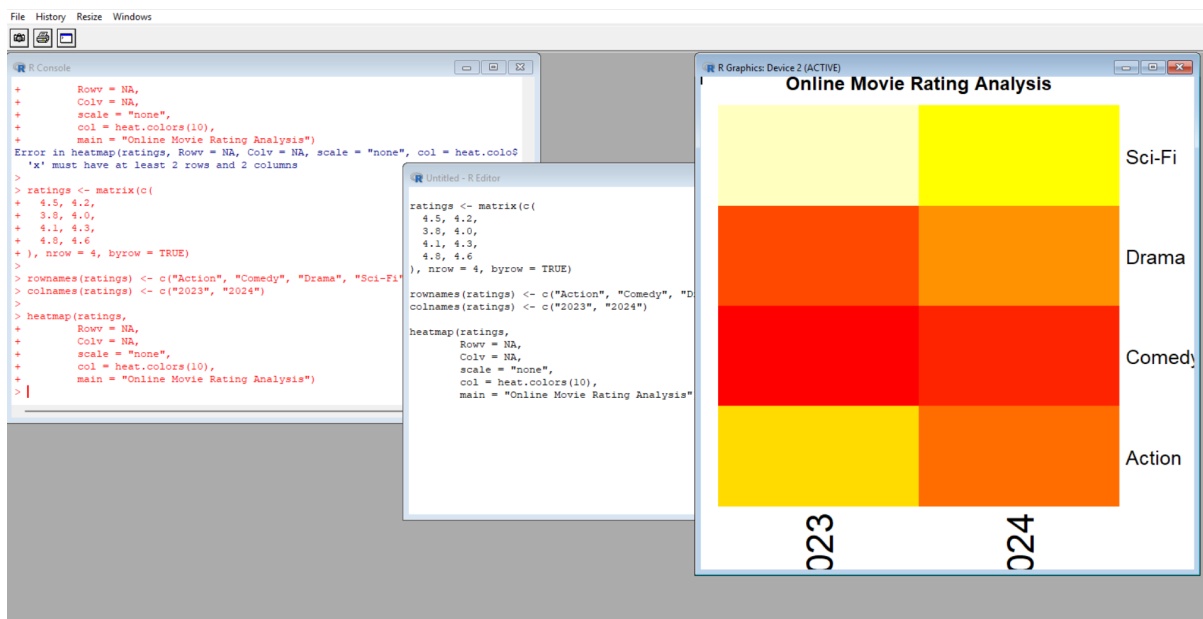


8: Online Movie Rating Analysis

```
ratings <- matrix(c(
  4.5, 4.2,
  3.8, 4.0,
  4.1, 4.3,
  4.8, 4.6
), nrow = 4, byrow = TRUE)

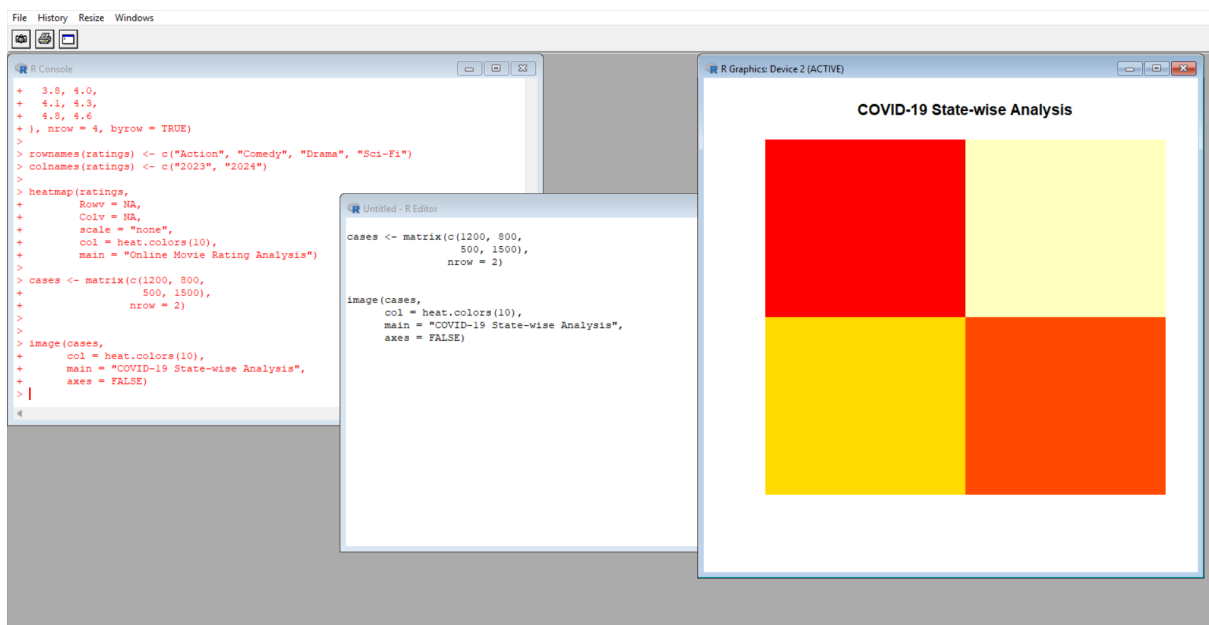
rownames(ratings) <- c("Action", "Comedy", "Drama", "Sci-Fi")
colnames(ratings) <- c("2023", "2024")

heatmap(ratings,
  Rowv = NA,
  Colv = NA,
  scale = "none",
  col = heat.colors(10),
  main = "Online Movie Rating Analysis")
```



9: COVID-19 State-wise Analysis

```
cases <- matrix(c(1200, 800,  
                 500, 1500),  
               nrow = 2)  
  
image(cases,  
      col = heat.colors(10),  
      main = "COVID-19 State-wise Analysis",  
      axes = FALSE)
```



10: Iris Flower Classification

```
plot(iris$Sepal.Length,  
     iris$Petal.Length,  
     col = as.numeric(iris$Species),  
     pch = as.numeric(iris$Species),  
     cex = iris$Sepal.Width/2,  
     xlab = "Sepal Length",  
     ylab = "Petal Length",  
     main = "Iris Flower Classification")
```

```
legend("topleft",  
      legend = levels(iris$Species),  
      col = 1:3,  
      pch = 1:3)
```

